ECML PKDD 2025

Porto.

LAMDA
Learning And Mining from DatA
http://lamda.nju.edu.cn

南京大学
NANJING UNIVERSITY 1902

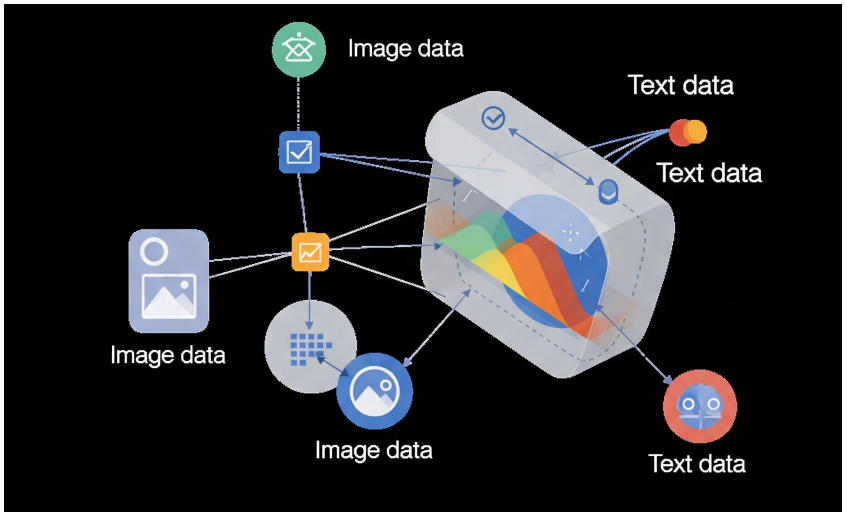
Voronoi Diagram Encoded Hashing

Yang Xu and Kai Ming Ting

State Key Laboratory for Novel Software Technology, Nanjing University, 210023, China

School of Artificial Intelligence, Nanjing University, 210023, China

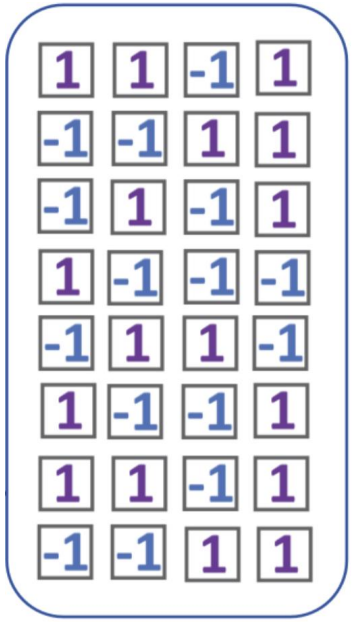
Motivation: The Goal of Hashing



High-dimensional data

Euclidean distance

L2H



Compact binary codes

Hamming distance

Two Doubts on the "Learning" in L2H

- **Doubt 1: Is complex learning necessary?**

  A recent study showed that a simple, no-learning LSH variant can achieve comparable accuracy to complex L2H methods, but much faster.

- **Doubt 2: Are we constrained by our tools?**
  - Existing L2H methods are limited to just three types of hash functions: **thresholding**, **hyperspheres**, and **hyperplanes**.
  - These functions require a complex, costly learning phase to work well.

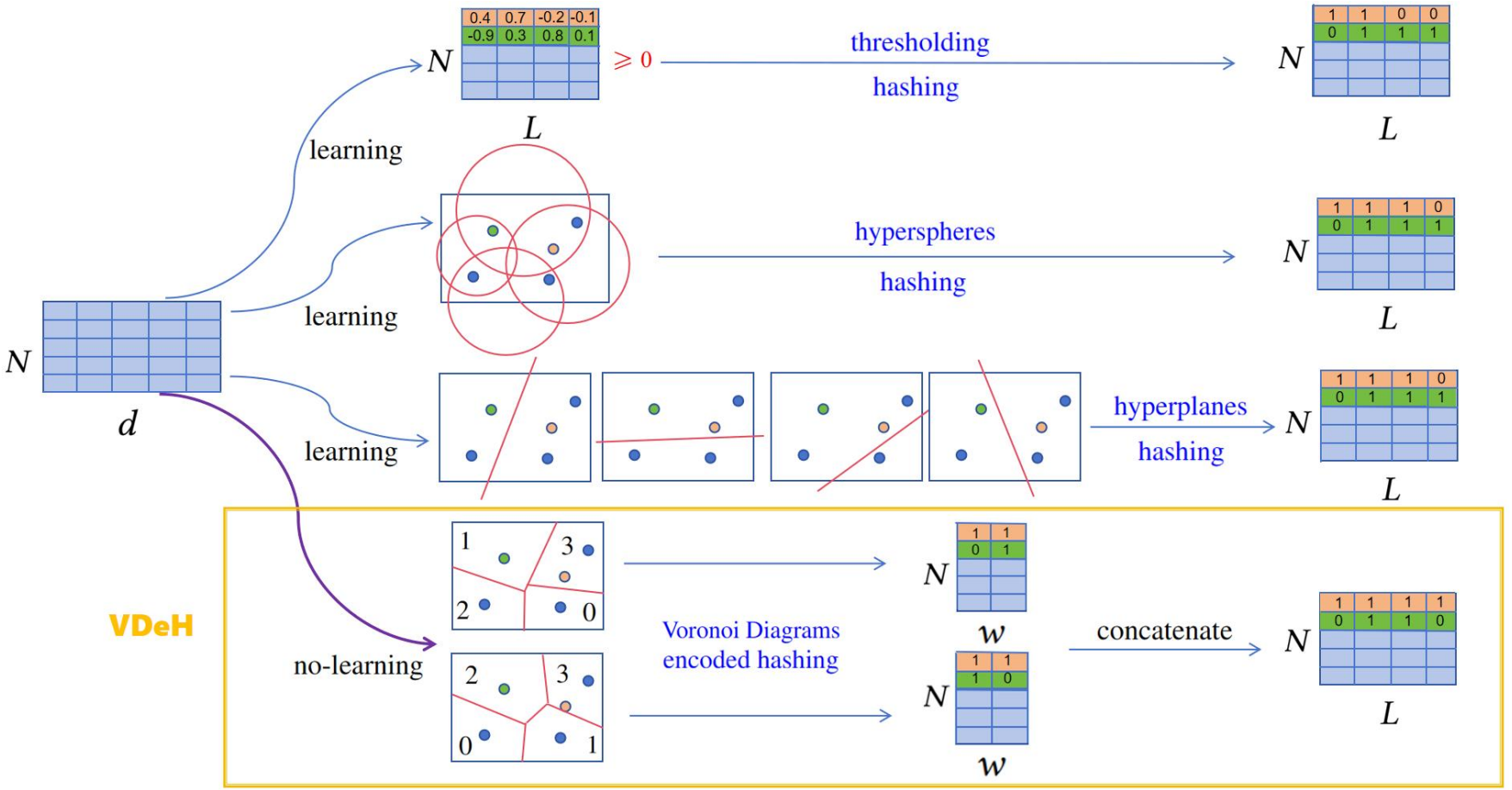
Effective Binary Hashing

- There are three widely claimed key properties for effective binary hashing:
  - **Full Space Coverage**: Every data point can be assigned a hash code. Avoids retrieval failures.
  - **Entropy Maximization**: Data is distributed evenly across all hash buckets. Maximizes information and avoids under-utilized codes.
  - **Bit Independence**: Each bit in the code provides unique information. Eliminates redundancy and improves efficiency.
- Existing methods use learning to enforce these properties.

Our Core Insight: Voronoi Diagrams

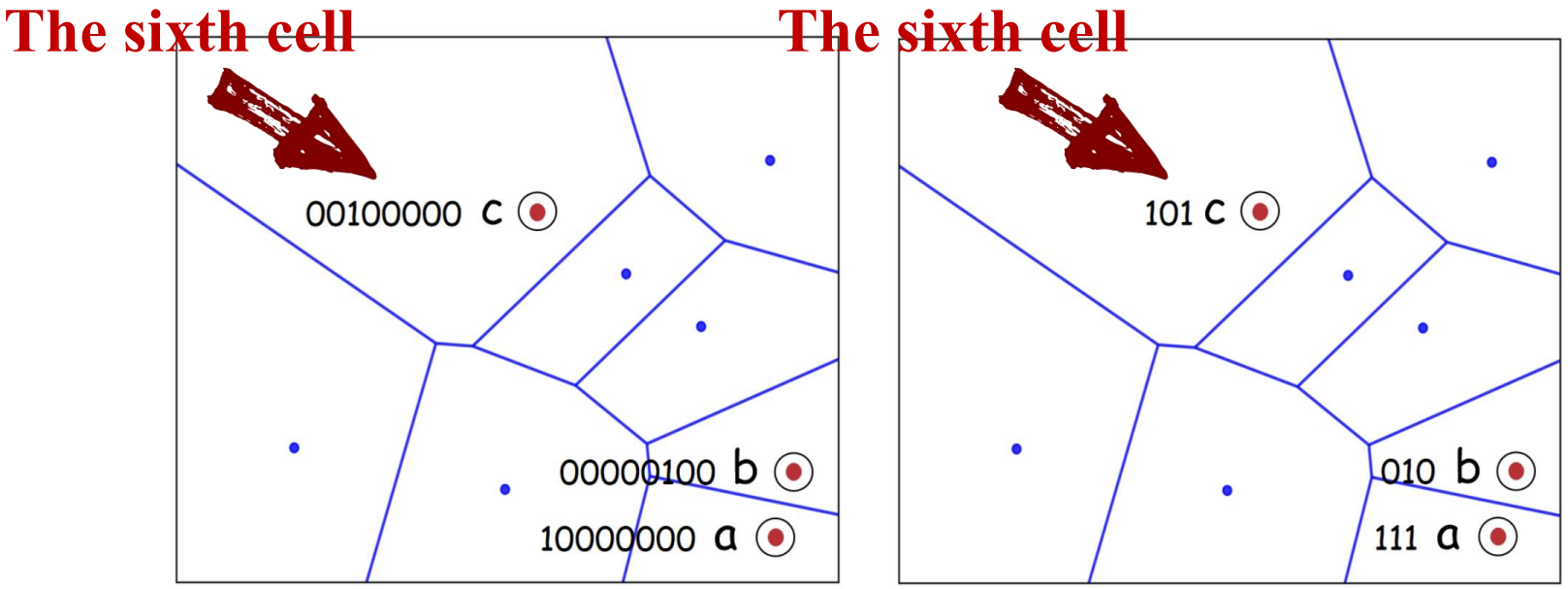
- **The Insight**: Voronoi diagrams inherently possess these three properties, **without any learning.**
  - **Natural Full Space Coverage**: The cells collectively partition and cover the entire space.
  - **Natural Entropy Maximization**: When generated from data samples, each cell tends to cover a similar number of points.
  - **Achievable Bit Independence**: It is feasible to transform a Voronoi diagram into a set of mutually independent hash functions.

Voronoi Diagram Encode Hashing



How VDeH Works: The Encoding is Key

- **Challenge:** A point x can only be in one Voronoi cell. A direct 1-of-k mapping (h_i(x)=1, others are 0) creates **dependent** bits.
- **Solution: Encoded Hashing**



(a) Sparse 8-bit embedding **(b) Dense 3-bit embedding**

The VDeH Algorithm

- **Step 1: Partition**

1. Given a dataset $\mathcal{X}$, we randomly sample ψ points to form a set $\mathcal{D}$, and generate the set of Voronoi diagram hash functions $H = \{h_1, h_2, \ldots, h_\psi\}$. According to Eq.(3), there is one and only one $h_i(\mathbf{x}) = 1$ for any point $\mathbf{x}$, and $\forall u \neq i, h_u(\mathbf{x}) = 0$.

- **Step 2: Locate & Index**

2. The 'raw' ψ-bit vector $[h_1(x), h_2(x), \ldots, h_\psi(x)]$, with $\psi \leq 2^w$, is encoded as a new binary vector with w bits:

$$b(x) = [e_1(x), e_2(x), \ldots, e_w(x)],$$

$$e_k(x) = L_{k-1} \bmod 2, \text{ where } L_0 = [\arg_{l \in [1,\psi]} h_l(x) = 1] - 1 \text{ and } L_k = \lfloor \frac{L_{k-1}}{2} \rfloor.$$

This encoding converts the 'raw' ψ-bit vector into its dense multi-bit binary representation, effectively converting an integer into a binary number.

The VDeH Algorithm

- **Step 3: Encode & Concatenate**

3. After repeating the above two steps L/w times, we concatenate all w-bit vectors $b(\mathbf{x})$ to formulate a L-bit code $\mathcal{B}(\mathbf{x}) = [b^1(\mathbf{x}), \ldots, b^{L/w}(\mathbf{x})]$ for any point $\mathbf{x}$.

Experimental Setup

- **Datasets**
  - **Images**: CIFAR-10 , GIST
  - **Text**: Nytimes , Kosarak (more challenging)
- **Baselines (9 methods)**
  - **Thresholding-based**: SH, CBE, ITQ, BP, SP
  - **Hyperspheres-based**: SpH
  - **Hyperplanes-based**: DSH, SELVE, rcLSH
- **Metrics**
  - Mean Average Precision (mAP)
  - Precision-Recall (PR) Curve

Results: Superior Accuracy

Dataset	bits	VD-based	HS-based	HP-based			Th-based				
		VDeH	SpH	rcLSH	DSH	SELVE	SH	CBE	ITQ	BP	SP
CIFAR-10	128	**.366**	.242	.283	.318	.199	.117	.315	.355	.344	.352
	256	**.438**	.289	.351	.372	.176	.153	.342	.377	.343	.370
	512	**.468**	.304	.392	.401	.167	.193	.396	.389	.339	.397
	1024	**.501**	.314	.412	.429	.169	.201	.401	.398	.332	.407
	2048	**.525**	.321	.432	.457	.169	.214	.407	.401	.335	.426
GIST	128	**.664**	.222	.582	.501	.430	.338	.617	.644	.645	.604
	256	**.714**	.223	.631	.580	.399	.454	.644	.650	.641	.614
	512	**.763**	.232	.651	.618	.402	.501	.656	.654	.634	.638
	1024	**.774**	.233	.659	.610	.399	.514	.657	.657	.642	.638
	2048	**.793**	.242	.687	.621	.398	.532	.662	.663	.644	.641
Nytimes	128	**.116**	.100	.017	.016	.003	.033	.022	.023	.023	.022
	256	**.165**	.110	.026	.033	.004	.055	.029	.029	.029	.027
	512	**.340**	.146	.028	.185	.004	.160	.026	.030	.026	.029
	1024	**.348**	.088	.103	.339	.006	.295	.108	.104	.008	.106
	2048	**.376**	.089	.113	.326	.008	.310	.100	.097	.008	.100
Kosarak	128	**.457**	.256	.235	.368	.203	.375	.255	.261	.254	.258
	256	**.603**	.296	.271	.494	.226	.512	.286	.293	.295	.304
	512	**.607**	.346	.308	.437	.279	.571	.301	.298	.346	.341
	1024	**.615**	.372	.307	.415	.289	.569	.300	.302	.351	.357
	2048	**.638**	.381	.313	.412	.300	.579	.309	.312	.359	.369

Results: Superior Accuracy

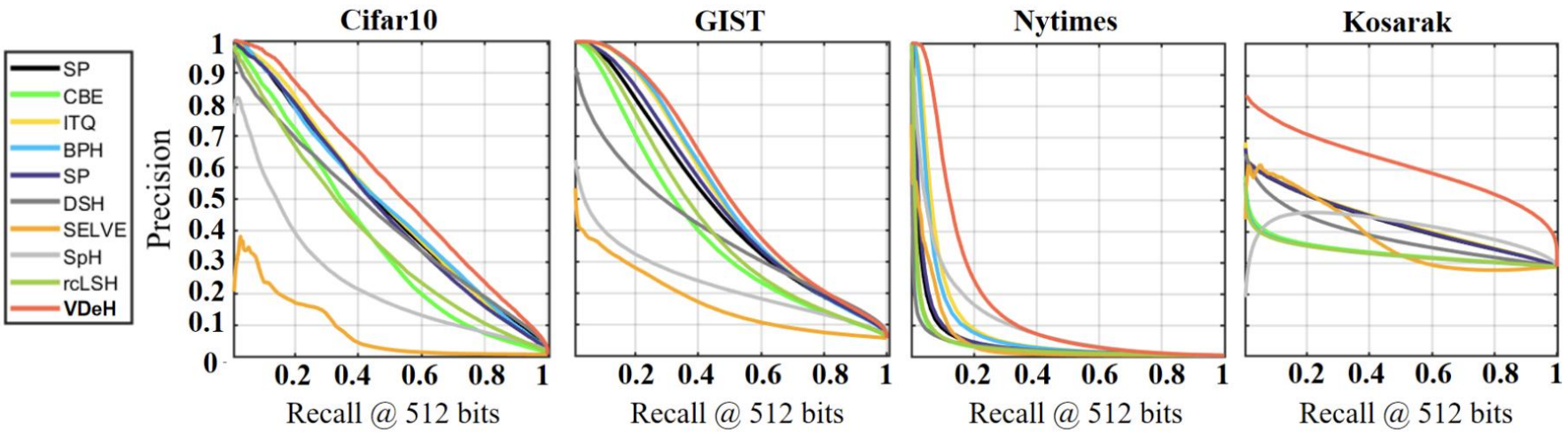


Fig. 2: PR curves on four datasets with the 512-bit binary codes

Results: Superior Accuracy

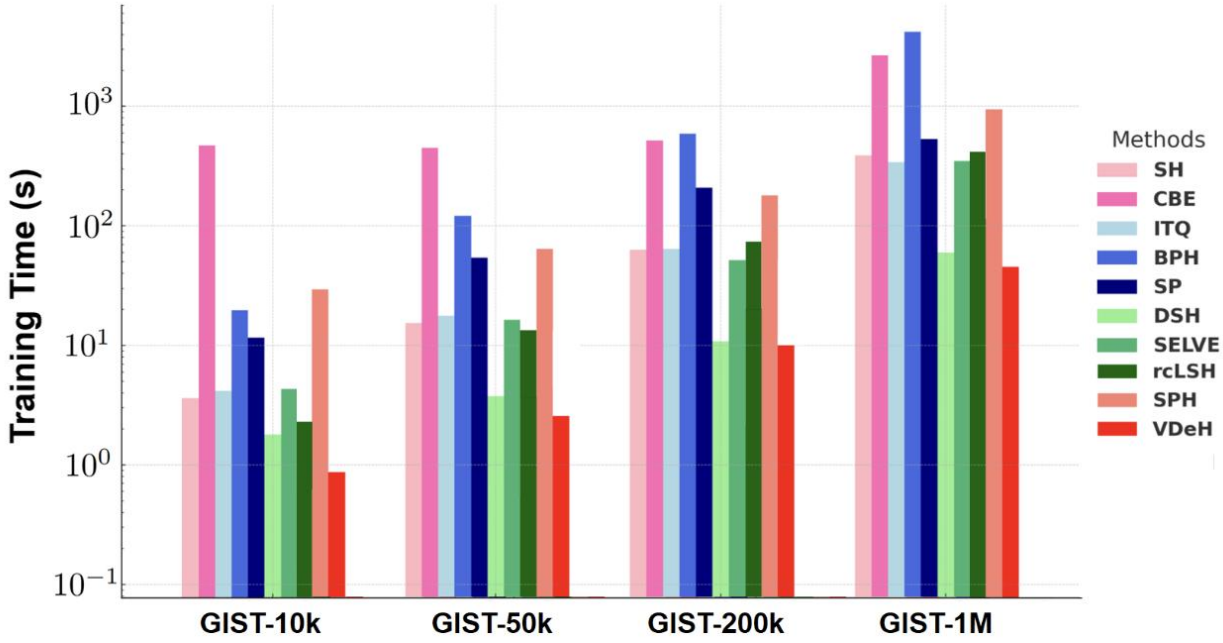


Fig. 4: Training time comparison on GIST dataset with the 512-bit binary codes.

Conclusion

- We introduced VDeH, a new no-learning data-dependent hashing method.
- **Key Contributions**:
  - **A New Perspective:** We identified that Voronoi diagrams naturally possess the three key properties that L2H methods struggle to learn.
  - **A Simple Method:** VDeH directly leverages these properties, creating a simple, fast, and data-dependent hashing scheme without complex learning.
  - **Demonstrated Superiority**: VDeH achieves state-of-the-art accuracy with significantly lower computational cost.


LAMDA
Learning And Mining from DatA
http://lamda.nju.edu.cn
南京大学
NANJING UNIVERSITY
1902

Thanks!

Q&A